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30 August 1985
SUPERSEDING
MIL-V-24272A(SHIPS)(IN PART)
24 September 1973
(See 6.7)

MILITARY SPECIFICATION

VALVE MANIFOLDS, QUIET DESIGN, HIGH PRESSURE GAS REDUCING

This specification is approved for use within the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the design, construction, test, and operating requirements for quiet design, high pressure air or nitrogen reducing valve manifolds.

1.2 Classification. Valve manifolds shall be of the following configurations, as specified (see 6.2.1):

Configuration 1-1 - One reducer, one relief (see figure 1).
Configuration 2-1 - Two reducers, one relief (see figure 2).
Configuration SP - Special configuration (see 6.2.1).

1.3 Pressure ratings. Valve manifolds shall have nominal inlet pressure ratings of 400, 1500, 3000 and 6000 pounds per square inch (lb/in^2), as specified (see 6.2.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362-5101 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 4820

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SPECIFICATIONS

FEDERAL

UU-P-268 - Paper, Kraft, Wrapping.
PPP-C-850 - Cushioning Material, Polystyrene, Expanded, Resilient (for Packaging Uses).
PPP-C-1120 - Cushioning Material, Uncompressed Bound Fiber for Packaging.

MILITARY

MIL-V-3 - Valves, Fittings, and Flanges (Except for Systems Indicated Herein); Packaging of.
MIL-S-901 - Shock Tests, H.I. (High-Impact); Shipboard Machinery, Equipment and Systems, Requirements for.
MIL-F-1183 - Fittings, Pipe, Cast Bronze, Silver-Brazing, General Specification for.
MIL-V-2961 - Valves, Pressure Reducing, for Gas Service (Sizes 1/4 to 2 inches IPS).
MIL-R-6130 - Rubber, Cellular, Chemically Blown.
MIL-F-8815 - Filter and Filter Elements, Fluid Pressure, Hydraulic Line, 15 Micron Absolute and 5 Micron Absolute, Type II Systems, General Specification for.
MIL-R-20092 - Rubber Sheets and Molded Shapes, Cellular, Synthetic, Open Cell (Foamed Latex).
MIL-V-22549 - Valves, Angle, Relief, for Gas and Oxygen Service (Sizes 2-Inches IPS and Below); Naval Shipboard.
MIL-V-22687 - Valves, Ball, Naval Shipboard, for Air, Nitrogen, Helium or Hydraulic Service (Sizes 2-1/2 Inches I.P.S. and Below).
MIL-V-24109 - Valves, Globe, Angle, Quick Change Cartridge Trim, High Pressure (H.P.) Hydraulic and Pneumatic (Sizes 1/8 - 1-1/4 Inches).
MIL-P-26514 - Polyurethane Foam, Rigid or Flexible for Packaging.

STANDARDS

FEDERAL

FED-STD-H28/2 - Unified Thread Form and Thread Series for Bolts, Screws, Nuts, Tapped Holes and General Applications.

MILITARY

MIL-STD-167-1 - Mechanical Vibrations of Shipboard Equipment (Type I - Environmental and Type II - Internally Excited).
MIL-STD-740 - Airborne and Structureborne Noise Measurements and Acceptance Criteria of Shipboard Equipment.
MS16142 - Boss, Gasket Seal Straight Thread Tube Fitting, Standard Dimensions for.

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2.1.2 Government drawings. The following Government drawings form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS

NAVAL SEA SYSTEMS COMMAND (NAVSEA)
NAVSHIPS 810-1385884 - Unions, Fittings and Adapters Butt and Socket Welding 6000 PSI, WOG and OXY (IPS).
NAVSHIPS 810-1385943 - Unions, Silver Brazing 3000 PSI, WOG, IPS for UT Inspection.
NAVSHIPS 810-1385946 - Unions, Bronze Silver Brazing, WOG for UT Inspection.

(Copies of specifications, standards and drawings required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order (see 6.2.1), a sample shall be subjected to first article inspection (see 4.3 and 6.3).

3.2 General description. Quiet design reducing valve manifolds shall incorporate all the components normally associated with a reducing station into a single assembly (body or block) requiring only one inlet and one outlet connection to the main flow path. The manifolds shall incorporate design features which enable them to meet the noise requirements specified herein when operating in the normal automatic regulating mode. The manifolds shall provide the following baseline features:

(a) Automatic means of regulating outlet pressure.
(b) Provision for attachment of a pressure gauge to monitor outlet pressure.
(c) Contamination (particulate) protection for the pressure regulators.
(d) Full-flow manual bypass.
(e) Provision for isolating regulators, gauges and filtration equipment.
(f) Nonisolatable relief valve protection for both automatic and manual operation.
(g) Suitable bleeds to permit depressurization of regulators, gauges, relief valves and filtration equipment.

3.3 Materials. Material requirements for subassemblies shall be as specified in the applicable component specifications referenced herein. Materials for all other parts, including manifold body, shall be selected to provide inherent corrosion resistance in a marine atmosphere without requiring

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painting, coating or plating. Materials for contacting parts shall be selected to minimize electrolytic corrosion and galling. Nonmetallic materials shall be compatible with ships air containing sea water vapor (100 percent humidity) and 2190 TEP lubricating oil. Manifold bodies shall be wrought or made from a wrought material.

3.3.1 Recovered materials. Unless otherwise specified herein, all equipment, material, and articles incorporated in the products covered by this specification shall be new and may be fabricated using materials produced from recovered materials to the maximum extent practicable without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products is allowed under this specification unless otherwise specifically specified.

3.3.2 Verification of material identification and control. The material covered by this specification will be used in critical shipboard systems. The use of incorrect material would create a high probability of failure resulting in serious personnel injury, loss of life, loss of vital shipboard systems or loss of the ship itself. The manifold has been designated as level I material and special control procedures are invoked to ensure receipt of correct material.

3.3.2.1 Materials which form the pressure boundary of the manifold shall be controlled by the level I requirements specified in 3.3.2.1.1 through 3.3.2.1.3.

3.3.2.1.1 Quantitative chemical and physical analysis of material traceable to heat identification are required for the following, which shall be designated on the drawings as "level I" or "I".

(a) Manifold body.
(b) Stems which penetrate the pressure boundary.
(c) Metallic pressure boundary parts.
(d) Threaded fasteners, 1/2 inch nominal and larger.

Certification documentation shall include, in addition to the material specification, the class, form, condition, grade, type and composition applicable to the material supplied (see 6.2.2).

3.3.2.1.2 Pressure seals, such as O-rings, are specifically excluded for level I material control requirements.

3.3.2.1.3 Questions regarding applicability of level I requirements shall be referred to the Naval Sea Systems Command (NAVSEA).

3.4 Design and construction.

3.4.1 Design concept. Sub-components shall be cartridge mounted in the manifold body and shall be readily replaceable without requiring removal of the manifold body from the line. The manifold body shall be of single-block construction such that flow paths between sub-components are provided by integral passages in the block and the joints normally required to connect

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sub-components into the flow path are eliminated. Sub-components shall emphasize ruggedness, reliability and simplicity. The complete manifold shall be as small, compact and light as possible, consistent with satisfactory performance, durability, reliability and safety. The manifold and all sub-components shall be designed so that engagement or disengagement of parts against any significant spring compression shall not be necessary. The manifold shall be completely repairable by using the tools specified in 3.4.6.2.

3.4.2 Design pressures. For design purposes, factors of 1.5 and 3 shall be used for proof and burst pressures. These factors shall be applied to the specified maximum operating pressure at the applicable manifold section, that is proof pressure of 1.5 times the nominal inlet pressure rating from the manifold inlet to the downstream cutout valve and 1.5 times the accumulation pressure of the relief valve from the downstream cutout valve to the manifold outlet.

3.4.3 End connections. The main-line inlet and outlet connections shall be in accordance with table I based on the design pressure requirements in accordance with 3.4.2. Union threadpiece adapters shall be secured to the manifold block with a straight thread and O-ring seal. The O-ring shall be retained in a groove on the outer diameter of the threadpiece adapter inboard of the adapter threads. Only the threadpiece dimensions of the documents specified in table I which are pertinent to the requirements of the straight thread and O-ring seal adapter configuration shall apply. The materials for the union tailpieces shall be as specified (see 6.2.1).

The end connections shall permit axial adjustment of the threadpieces to expedite proper installation of the manifold. The range over which each connection can be adjusted shall be specified in table II. A positive and permanent means shall be incorporated to insure that threadpieces are not backed out beyond their minimum engagement position. Size of end connections shall be as specified (see 6.2.1).

TABLE I. End connections.

Pressure rating (lb/in^2)	Applicable document
400	MIL-F-1183 (O-ring type)
1500	810-1385946
3000	810-1385943
6000	810-1385884

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TABLE II. Adjustable range of end connections.

Connection size nominal pipe size (nps)	Minimum adjustment range (inch)
1/8, 1/4 and 3/8	3/8
1/2, 3/4, 1 and 1-1/4	1/2
1-1/2, 2 and 2-1/2	3/4

3.4.4 Mounting. The body of the manifold shall have provision for either bottom or back mounting to the ship structure as specified (see 6.2.1). Sub-components used in the manifold assembly shall be easily accessible for operation, adjustments, service or removal from the manifold, with the manifold mounted to the ship structure.

3.4.5 Threads. Threads shall be in accordance with FED-STD-H28/2. Pipe threads shall not be used. Where necessary, provisions shall be incorporated to prevent the accidental loosening of threaded parts. The design shall be such that standard wrenches may be used on all external bolting. Lock-wire shall not be used. Low efficiency threads with small pitch to diameter ratios shall not be used.

3.4.6 Interchangeability. The entire manifold, including sub-components, shall incorporate complete parts interchangeability. Each part (see 3.4.6.1) shall have part number identity, and shall be replaceable from stock or the manufacturer on a nonselective and random basis. Parts having the same manufacturer's part number shall be directly interchangeable with each other with respect to installation (physical) and performance (function). Physical interchangeability shall be in accordance with the following definition: Interchangeable assemblies, components and parts are those which are capable of being readily installed, removed or replaced without alteration, misalignment, or damage to parts being installed or to adjoining parts. No fabrication operations such as cutting, filing, drilling, reaming, hammering, bending, prying or forcing shall be required.

3.4.6.1 Parts. Nonmetallic elements, including but not limited to, seat rings, poppet seat inserts, cushions, and O-rings shall be treated as separately identified and readily replaceable parts.

3.4.6.2 Tools. Any special tools required for maintenance of the manifold shall be supplied with the manifold. Special tools are defined as those tools not listed in the Federal Supply Catalog (copies of this catalog may be consulted in the office of the Defense Contract Administration Services Management Area (DCASMA)).

3.4.7 O-ring locations. As far as practical, O-rings in face-seal applications shall have the grooves located in the lower members to simplify assembly.

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3.4.8 Gauge connection. The manifold shall be provided with a 1/4-inch gauge connection port in accordance with MS16142 to permit attachment of a pressure gauge for sensing the regulated outlet pressure downstream of the downstream cutout valve. The gauge connection shall be located to accurately measure pressure at the manifold outlet connection when using either the pressure regulating valve or the manual bypass valve to control flow. A gauge isolation valve shall be provided in the manifold.

3.4.9 Bleed valves. Bleed valves shall be provided to insure depressurization of piping and components between the upstream and downstream cutout valves. A gauge bleed valve shall be provided to permit depressurization and removal for testing of the pressure gauge. The gauge bleed valve may be omitted on application where a gauge bleed valve will be provided external to the manifold. Whether a gauge bleed valve is required to be supplied as part of the manifold shall be specified (see 6.2.1).

3.4.10 Identification plates and markings. Identification plates made of corrosion-resisting steel, brass or anodized aluminum, shall be permanently fastened to a low-stress portion of the valve manifold body, and shall contain the following data or a space therefor:

(a) Manufacturer's name.
(b) Capacity rating of manifold in standard cubic feet per minute (std ft^3/min) at a given set of upstream and downstream pressure conditions.
(c) Nominal inlet pressure rating.
(d) Range of operating inlet pressures.
(e) Outlet pressure range of manifold.
(f) Manufacturer's drawing and CD sheet numbers.
(g) Manufacturer's model and part number for manifold.
(h) Space for 9 digit CID number.

3.4.10.1 Each major sub-component (such as regulators, relief valves, manual valves, differential indicators, and filters) shall also have an identification plate. This plate shall contain the following data:

(a) Manufacturer's name.
(b) Manufacturer's drawing number.
(c) Manufacturer's part number.

Where insufficient space results from the cartridge configuration, these plates may be mounted to the manifold block immediately adjacent to the sub-component, or the information may be electro-etched on an outside visible surface of the sub-component.

3.4.10.2 The manifold shall have an identification plate with the schematic diagram of the manifold and detailed operating instructions. This plate may be supplied as a separate unit to be mounted off the manifold, if space does not permit direct mounting on the manifold body. The schemetic diagram shall follow the basic format and symbols shown on figures 1 and 2.

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3.4.11 Sub-components.

3.4.11.1 Reducing valves. The requirements and definitions for pressure reducing valves incorporated in the manifold shall be in accordance with MIL-V-2961, except as modified in 3.4.11.1.1 through 3.4.11.1.4.4 and as otherwise specified herein.

3.4.11.1.1 The parallel reducing valves on configuration 2-1 manifolds shall be identical from a design and construction standpoint. Each shall be sized to pass full rated flow of the manifold.

3.4.11.1.2 Reducing valves shall be of a simple pattern with the fewest dynamic seals compatible with the accuracy required. Configuration shall be a cartridge type, compatible with installation and operation within the manifold block.

3.4.11.1.3 Where type II (gas-dome loaded) regulators in accordance with MIL-V-2961 are used, the dome shall be a high-integrity design, maintaining dome charge without adjustment or recharge more frequently than once a year for the manifold to remain within its specified performance envelope. Upstream pressure shall be utilized to establish dome load. The line to accomplish this, and all other pressure lines, shall be internally ported within the manifold structure. Dome loading shall be accomplished by two simple metal-seated valves installed in series, with a bleed-off valve between. Only the single valve penetrating the dome shall affect dome integrity. The valves shall be operable by a standard-size hex wrench or other suitable means. There shall be no external leakage past this arrangement at any time. Flow from the bleed-off valve shall be ported in such a way that it does not impinge directly on the operator, cause excessive noise, or potentially lead to ice formation within the dome loading circuit.

3.4.11.1.4 Quiet-flow regulating element (QFRE). The design employed to achieve a low-noise pressure loss shall maintain maximum operational reliability by avoiding features such as flow passage sizes and patterns susceptible to clogging. The level of filtration specified in 3.4.11.5 shall be adequate to protect the element from blockage or clogging.

3.4.11.1.4.1 The QFRE shall be capable of withstanding installation and operational stresses. Cleaning shall not be required more often than yearly. The QFRE shall be capable of being quickly cleaned aboard ship, either by back-flush or using ultrasonic equipment. Any special fixtures necessary for cleaning of the QFRE shall be supplied with the manifold in accordance with 3.4.14.

3.4.11.1.4.2 The structure of the QFRE shall be of a solid or continuous nature, not susceptible to media migration.

3.4.11.1.4.3 The QFRE, and its interrelationship with the valving element and the remainder of the manifold, shall prevent ice formation from blocking flow at any operating condition within the required range of the manifold and with air having a moisture content within the limits specified in 3.4.12. Where close metal-to-metal clearances are employed to accomplish the above, construction shall still assure reliable functioning without susceptibility to binding, erosion or excessive wear.

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3.4.11.1.4.4 No special start-up or operating procedures shall be required to prevent ice blockage within the manifold.

3.4.11.1.4.5 Pressure reversal. Regulators shall withstand, without damage, a condition where maximum set pressure is trapped between the manifold outlet valve and the regulator seat. This can occur where maximum set pressure exists at the regulator outlet, and its loading element is deactivated (that is, if it is a spring loaded regulator, the set spring adjustment is backed off fully, and if a gas-dome loaded regulator, the dome charge is vented off completely).

3.4.11.2 Relief valves. The requirements and definitions for pressure relief valves incorporated in the manifold shall be in accordance with MIL-V-22549, except as modified below and as otherwise specified herein. Setting and performance requirements shall be as specified in 3.5. Relief valves shall be direct spring-loaded, as specified in MIL-V-22549, and shall incorporate no sliding seals of any kind.

3.4.11.3 Hand valves. The main-line hand valves for regulator isolation and by-pass shall incorporate a nonmetallic seating feature for tight shutoff. Maximum permissible operating torques for all hand valves shall be in accordance with MIL-V-24109. The manual regulator by-pass valve shall be sized to pass full-rated flow of the manifold and with a plug form permitting precise manual control of the outlet pressure at all flow rates within the range of the manifold. Hand valves shall be capable of full performance and tight shutoff when pressurized in either direction.

3.4.11.3.1 Manual valves shall have plug forms to permit gradual pressurization and depressurization. Manual isolation valves shall be capable of being operated in a manner capable of limiting the rate of downstream pressure buildup in a depressurized volume, with maximum pressure upstream, to 200 lb/in^2 per second. Downstream volumes for this pressurization rate requirement shall be taken as the applicable manifold volumes. For design and test purposes, a stop valve not more than 10 diameters downstream of the manifold outlet connection shall be assumed.

3.4.11.4 Soft-seating inserts. Soft-seating inserts used within the manifold shall be protected from direct flow impingement, excessive loading and extrusion, or any other effect jeopardizing their useful life. Inserts shall be replaceable on an interchangeability basis as specified in 3.4.6. Inserts shall be of the simplest practical configuration to allow emergency replacement manufacture aboard ship.

3.4.11.5 Filters. Performance, construction and material requirements for the filters shall be in accordance with 3.4.11.5.1 through 3.4.11.5.4.

3.4.11.5.1 Particulate removal. Particulate removal requirements shall be as follows:

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Manifold nominal pressure rating (lb/in^2)	Particulate removal (micrometers) nominal (98 percent removal)/ absolute (100 percent removal)
400	20/50
1500	5/18
3000	5/18
6000	5/18

3.4.11.5.2 Pressure drop requirements. Filters shall be sized so that when clean, the pressure drop across the filter element, at the rated flow of the valve manifold, does not exceed the following:

Manifold nominal pressure rating (lb/in^2)	Pressure drop (lb/in^2)
400	10
1500	40
3000	75
6000	150

3.4.11.5.3 Differential indicator. Filters shall be provided with a nonelectrical differential pressure indicator which will visually warn by raising a red indicator when the pressure drop across the filter element exceeds the allowable clean pressure drop, at rated flow, by 50 percent. The indicator shall be actuated by sensing the pressure drop across the filter element only. The red indicator, once actuated, shall remain up until manually reset.

3.4.11.5.4 Filter element. The filter element shall be made of nickel-copper alloy or austenitic corrosion-resisting steel. The element shall withstand a differential pressure equal to the maximum inlet pressure and a surge flow condition imposed by rapidly pressurizing and depressurizing the element using maximum inlet pressure. Cleaning and reuse of the element shall be permitted, and once cleaned, shall be capable of meeting the requirements for a new element. The element shall not be affected by immersion in water.

3.4.12 Environmental conditions. Manifold performance shall not be adversely affected by the following line and ambient conditions:

(a) Line: Air or nitrogen moisture content between the limits of plus 20 to minus 60 degrees Fahrenheit (°F) dewpoint at 4500 lb/in^2.

(b) Ambient:

(1) Salt spray: Exposure to atmosphere containing salt laden moisture.
(2) Vibration: Environmental vibration as specified in type I of MIL-STD-167-1.
(3) Shock: Mechanical shock as specified in grade A, class I of MIL-S-901.
(4) Temperature: 40 to 120°F.

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3.4.13 Cartridge orientation. Where specific orientation of cartridge assemblies within the manifold block is required, the index means employed shall not be susceptible to being inadvertently overridden, nor to jamming or otherwise damaging parts of the manifold if incorrectly installed.

3.4.14 Test and cleaning fixture. Where specified (see 6.2.1), a combination test and cleaning fixture shall be provided. This fixture shall have the same pressure rating as the inlet of the manifold body, and shall be made from noncorrosive material. Level I material identification and control requirements do not apply to this fixture. The inlet and outlet shall be for the intended purpose of this fixture as defined herein. This fixture shall be capable of accommodating the QFRE. It shall be capable of backflush cleaning the QFRE, plus verifying flow capacity of cleaned or new replacement QFRE at pressure and flow conditions available at a tender or Naval land-based maintenance facility. The applicability of each fixture shall be clearly marked. An instruction plate shall be permanently attached to each fixture which provides cleaning and flow verification procedures, including all necessary flow and pressure parameters, and specific reference to applicable technical manual paragraphs containing this information.

3.5 Performance.

3.5.1 Capacity. The flow capacity required shall be specified (see 6.2.1), in std ft^3/min, for a given set of upstream and downstream pressure conditions. These shall be based on the minimum inlet pressure and the highest reduced pressure setting under which the manifold will be required to operate. The range of upstream and set pressures over which the manifold is required to operate and meet the specified performance requirements shall also be specified (see 6.2.1). The manifold shall meet the specified capacity requirements, or any intermediate capacity requirement, and shall operate without hunting, chattering or excessive noise or vibration, under all specified conditions.

3.5.2 Accuracy of regulation. Unless otherwise specified (see 6.2.1), the regulated pressure shall be maintained within the limits specified on figure 3 under all flow and inlet pressure conditions specified. Figure 3 shows the manifold set pressure versus the maximum allowable droop in regulated pressure. The set pressure shall be chosen as the lock-up pressure for the purposes of this specification. The droop is defined as the maximum variation in regulated pressure which occurs when the manifold is subjected (without adjustment or resetting) to all possible combinations of inlet pressure and flow within the specified limits. The regulated, outlet or downstream pressure referred to in this specification shall be considered to be the pressure measured at the manifold gauge connection. The required set or lock-up pressure shall be determined by adding the droop to the minimum outlet pressure required by the application.

3.5.3 Overpressure protection. The relief valves shall be sized to pass the wide open capacity of one regulating valve, or the wide open capacity of the manual regulator by-pass valve, whichever is greater, and limit the downstream pressure to the accumulation or rated flow pressure shown on figure 4. The relief valve setting shall be based on the nominal lock-up pressure of the manifold. The particular cracking pressure of the relief valves is for reference only. Unless otherwise specified (see 6.2.1), the relief valves shall operate

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within the accumulation and reseat envelope specified on figure 4. The manufacturer shall furnish the cracking pressure, with tolerance, required to meet the envelope requirements as shown on figure 4, in order to permit correctly setting and checking the relief valve settings when the manifolds are installed aboard ship. In all cases, regulator and relief valve characteristics shall be compatible, without overlap.

3.5.4 Seat tightness. Manual valves shall be bubble-tight in both directions when closed with a torque not exceeding that specified in MIL-V-24109 or MIL-V-22687, as applicable (bubble-tight being defined as no visible leakage over a 5 minute period with the unpressurized side, or a line from the unpressurized side, submerged in water or with bubble-fluid applied). Unless otherwise specified (see 6.2.1), internal leakage from the inlet to outlet side of the pressure regulating valves and from the inlet to discharge side of the pressure relief valves, at lock-up pressure, shall also meet the above defined bubble-tight criteria. Where necessary, leakage measurement shall start after temperature stabilization.

3.5.5 External leakage. There shall be no visible external leakage over a 5-minute period, using either bubble-fluid or submerging the manifold in water.

3.5.6 Noise. The manifolds shall not exceed the airborne noise criteria specified for equipment grade D in MIL-STD-740, when measured in accordance with the sound pressure level measurement procedures of MIL-STD-740, and shall not exceed the structureborne noise criteria specified on figure 5.

3.6 Technical data. The contractor shall prepare drawings and technical manuals in accordance with data ordering documents specified (see 6.2.2) and as specified in 3.6.1 and 3.6.2.

3.6.1 Drawings. In addition to the general requirements specified (see 6.2.2), the following shall be included as part of the drawings content.

(a) Accurately scaled assembly drawings of all components which clearly depict design and construction.
(b) Bill of material listing specification, grade, condition and any other data required to fully identify the properties of the materials proposed.
(c) Complete production details of all replenishment parts.
(d) In cases where standard military parts are or can be employed, these shall be appropriately identified. For example, O-rings and back-up rings shall be identified by MS number. Parts which are not standard military parts, but which have standard commercial identification shall be so identified.
(e) Outline dimensions, disassembly space, location and size of end connections and mounts.
(f) Estimated weight and limitations on installation center of gravity (vertical, longitudinal, and transverse).
(g) Performance characteristics of the equipment proposed under the specified conditions.
(h) Recommended assembly torques, or equivalent procedures for making up all joints and threaded assemblies.

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(i) Complete tabulated data on all elastomeric materials used in the manifold, including identifications, material and size designations, shore hardness and any other data necessary to fully identify.
(j) Calculated average fluid velocity through various portions of the valve. These shall include the inlet and outlet ports and, at minimum, enough sections within the valve to cover any velocity change in excess of 25 percent.
(k) Engineering parameters of the QFRE, including:

(1) Data on maximum particulate sizes that can be passed without clogging.
(2) Brief discussion or rationale on the mechanics by which pressure breakdown is achieved without generating the noise level normally associated with orificial throttling.
(3) Method of fabrication: Configuration and process of fabrication shall be described in sufficient detail to illustrate that adequate structural strength and freedom from flow passage obstruction is assured during production.

(l) Calculations verifying compliance with the flow passage and overall flow-rate consistency requirements specified in 3.4.11.1.4. Required dimensional inspection shall be detailed.
(m) Sub-contracted parts - For all sub-contracted parts, reference to the manufacturer's identification shall be provided. When more than a single sub-contractor can apply, identify at least two major sources.

3.6.1.1 In addition to the general requirements specified (see 6.2.2), the following certification data sheet information shall be included:

(a) General:
(1) Ship identification.
(2) Applicable assembly drawing number.
(3) Applicable manual number.
(4) CID (APL) number.
(5) Application description.
(6) Manifold description.

(b) Regulation:
(1) Set pressure and adjustable range.
(2) Required maximum capacity over specified range of operating conditions.
(3) Rated maximum capacity over specified range of operating conditions.
(4) Specified operating conditions - range of inlet pressures and required accuracy of regulation.
(5) Maximum fail-open capacity (for purposes of relief valve sizing) of regulating valve.
(6) Maximum wide-open capacity (for purposes of relief valve sizing) of manual by-pass valve.

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(c) Overpressure protection:
  (1) Set pressure and adjustable range.
  (2) Rated capacity - accumulation.
  (3) Blowdown.

(d) Filtration:
  (1) Clean pressure drop at manifold rated capacity.
  (2) Differential indicator setting.

3.6.1.2 Manifold drawing layout. Overall cross-sectional views of the entire manifold shall not be required. Assembly drawings and internal parts details, as specified in 3.6.1 and 3.6.1.1, however, shall be provided for each subassembly. The internal parts details shall be included either with the associated assembly view or on a second sheet bearing the same drawing number. Individual parts sheets shall not be used. An 8-1/2 by 11 inches index sheet (see figure 6) shall reference these subassemblies so as to completely define the overall manifold. Any variation, regardless of how slight, shall be covered by a new index sheet and model designation. The model designation shall uniquely define each manifold. For example, even if only an O-ring material change, or tailpiece variation, applies, a new model designation must be assigned (for example, manufacturer's name, PMR11GH-2, and so forth). In addition to the subassembly drawings, a drawing of the manifold block (not necessarily in cross-section) shall be required. This shall include only the block plus such permanently affixed parts as identification plates and studs. Capscrews threading into the block are preferably included in the drawings of the subassemblies which they attach.

3.6.2 Technical manuals. In addition to the general requirements specified (see 6.2.2) the following shall be included as part of the manual contents.

(a) The engineering drawings for the valve (including certification data sheet). These drawings shall be supplemented by additional illustrations where necessary to adequately illustrate operation and maintenance. These additional illustrations may consist of blowouts, partial or full sections, and so forth, and may eliminate extraneous lines and details to clarify the interaction of parts.
(b) Table listing wrench sizes and assembly torques (or other equivalent procedures) for making up all joints and threaded assemblies.
(c) Detailed disassembly and reassembly procedures. In addition to providing procedures for the complete disassembly and reassembly of the equipment, maintenance and troubleshooting sections shall contain, or refer to, only the limited disassembly and reassembly required to accomplish each particular operation. This is intended to reduce the possibility of unnecessary disassembly and unnecessary disturbance of adjustments when performing specific or limited maintenance or troubleshooting operations.
(d) Adjustment procedures.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 First article inspection. First article inspection shall consist of the examination of 4.5.1 and the tests specified in 4.6.1 through 4.6.19. The manifold assembly submitted for first article inspection shall consist of the following items:

(a) Internal replaceable trim; QFRE, poppets, springs, filter elements, and so forth - one complete set per manifold.
(b) Nonmetallic parts: O-rings, seating inserts, gaskets, and so forth - two complete sets per manifold.
(c) Special tools required for installation, operation and maintenance - one complete set per manifold.
(d) Instruction book or preliminary technical manual for installation, operation and maintenance - one per manifold.
(e) Drawings of test manifold in accordance with 3.6.1.

4.3.1 First article test procedure. Procedures covering all tests and inspections to be conducted for first article shall be prepared in accordance with the data ordering documents specified (see 6.2.2).

4.3.2 First article inspection report. A first article inspection report shall be prepared in accordance with the data ordering documents specified (see 6.2.2).

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4.4 Quality conformance inspection. Each production manifold shall be examined as specified in 4.5.2 and tested as specified in 4.6.1, 4.6.4, 4.6.8, 4.6.9, 4.6.10, 4.6.11 and 4.6.14. Production manifolds selected in accordance with the sampling plan (see appendix) shall be tested as specified in 4.6.19. Any discrepancies or failures shall be corrected prior to acceptance. Test reports shall be prepared in accordance with the data ordering document included in the contract or order (see 6.2.2).

4.5 Examination.

4.5.1 First article examination. Upon receipt of the first article, it shall be disassembled and visually examined to determine conformance with the requirements of this specification. On reassembly, all nonmetallic parts, except the end connection O-rings, shall be replaced with spares, randomly selected from those provided with the manifold. This shall be accomplished entirely by test personnel using only the technical manual supplied with the manifold for guidance.

4.5.2 Quality conformance examination. The manifold shall be examined to determine conformance with the requirements as specified in 3.2, 3.3 and 3.4. Examination shall be limited to external examination without disassembly.

4.6 Tests.

4.6.1 Proof. Each portion of the manifold (that is upstream and downstream of the regulator cutout valve) shall be proof tested at the applicable pressure specified in 3.4.2. Pressure shall be applied for 2 minutes, depressurized and pressure applied again for 2 minutes. Water, nitrogen or air may be used. There shall be no external leakage, permanent distortion or structural failure.

4.6.2 High-impact shock. The manifold shall be subjected to and meet the high-impact shock tests for grade A, class I as specified in MIL-S-901 pressurized with water, air or nitrogen. There shall be no structural damage to the manifold or any sub-components. There shall be no degradation to the performance capability of the manifold.

4.6.3 Vibration. The manifold shall be vibration tested in accordance with type I of MIL-STD-167-1 pressurized with water, air or nitrogen. There shall be no resonance in the range of frequency tested. There shall be no structural damage or degradation to the performance capability of the manifold.

4.6.4 Accuracy of regulation. The manifold shall be tested for accuracy of regulation at each inlet pressure and set pressure combination specified below. Flow rate at each condition shall be varied over the full operating range as specified in 6.2.1.

Condition	Inlet pressure	Set pressure
A	Maximum	Minimum
B	Minimum	Minimum
C	Minimum	Mid-point
D	Maximum	Mid-point
E	Maximum	Maximum
F	Minimum	Maximum

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During each sequence (conditions A-B, C-D, and E-F), no alteration shall be made to the set pressure adjustment, or any other portion of the manifold, and regulated pressure delivered shall be maintained within the band as shown on figure 3. There shall be no instability or other evidence of unsatisfactory operation of the manifold during these tests.

4.6.5 Accumulation. The relief valve shall be tested for accumulation capacity under both the condition of a full-open regulating valve and a full-open by-pass valve. A maximum rated inlet pressure shall be maintained at the manifold inlet during this test. For the regulator portion, close by-pass valve and block, or otherwise modify, regulating valve to the wide-open fail condition. For the by-pass valve portion, the regulator shall be isolated and the by-pass valve fully opened. Under either condition, downstream pressure rise shall not exceed the allowable shown on figure 4.

4.6.6 Manual by-pass valve functional. Minimum inlet pressure shall be maintained at manifold inlet. Regulators shall be isolated. By-pass valve shall be open until highest set pressure is maintained at maximum flow demand. Increase inlet pressure to maximum, then vary flow demand throughout the flow range of the manifold and verify ability of the by-pass valve to permit maintaining delivered pressure at set point in smooth accurate manner with good resolution. Operating torques shall not exceed those specified in 3.4.11.3.

4.6.6.1 Manual by-pass valve cycling. With maximum pressure rating upstream, the manual by-pass valve shall be cycled 1000 times. There shall be no degradation to performance or evidence of damage or excessive wear.

4.6.7 Hand-valve pressurization rate and functional. With maximum pressure rating upstream, and a depressurized downstream volume as specified in 3.4.11.3.1, each hand-operated valve shall be operated in accordance with the operating instructions affixed to the manifold to demonstrate its ability to meet the pressure buildup rate as specified in 3.4.11.3.1. Operating torques shall not exceed those specified in 3.4.11.3.

4.6.8 Relief valve operating. Pressure at the relief valve inlet shall be increased until the relief valve lifts. Pressure shall then be decreased until the relief valve reseats. This sequence shall be repeated 10 times. At all times, lift pressure shall not deviate by more than 2 lb/in^2, or 2 percent, whichever is greater, and reseat shall be within the envelope shown on figure 4. There shall be no evidence of operation instability or damage to the relief valve.

4.6.9 Seat tightness. Hand valves shall be seated with a torque not exceeding that specified in MIL-V-24109 or MIL-V-22687, as applicable. Air, at the highest operating pressure each valve is subjected to in service, shall be used for leakage tests, using bubble fluid or immersing the outlet, or a line from the outlet, under water. The hand valves used for isolation at the outlet of each regulating valve shall also be tested in the outlet-to-inlet direction, applying a pressure at the outlet of the valve equal to the maximum outlet-section rating of the manifold. Regulating valves shall be tested with an inlet pressure equal to the manifold rating. Relief valves shall be tested at the reseat pressure applicable to their setting. None of the valves shall show visible evidence of leakage over a 5-minute period. The following procedure

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shall be used to check regulating valve seat tightness: The valve's exact set-point shall be determined by the accuracy of regulation test. It shall then be isolated downstream. The dead-ended volume shall be tight and shall not exceed 10 diameters of downstream pipe. It shall be monitored with bubble-fluid to assure tightness. With full-rated inlet pressure, outlet pressure shall not rise above initial set lock-up checking over a 1-hour period. Initial lock-up shall not exceed that shown on figure 3 for the previously determined set-point.

4.6.10 External leakage. Air shall be applied at the appropriate maximum rating to each section of the manifold. Leakage shall be checked using bubble fluid or submerging manifold in water. There shall be no visible leakage over a 5-minute period.

4.6.11 Filter element bubble point. Test filter element for integrity and micrometer rating using the bubble-point test in accordance with MIL-F-8815. Initial bubble-point valve shall be not less than 4.5 inches of water for 400 lb/in^2 rated manifolds and 13 inches of water for 1500, 3000 and 6000 lb/in^2 rated manifolds. Where the type of filter used is not compatible with this test method, alternative procedures may be used where authorized by NAVSEA.

4.6.12 Filter element strength. Apply a pressure differential equal to the maximum pressure rating of the manifold across the filter element for 2 minutes. There shall be no structural damage to the filter element or media migration.

4.6.13 Filter element flow-surge. Apply a flow-surge across the filter element by pressuring the inlet side with a pressure equal to the maximum rating of the manifold. This shall be followed by depressurization to atmospheric pressure. The pressurization and depressurization phases shall each occur in 1 second or less. Repeat until 100 pressurization-depressurization cycles are completed. There shall be no structural damage to the filter element or media migration.

4.6.14 Filter operational. The filter shall be tested to determine the clean pressure drop across the element at rated flow. Pressure drop shall not exceed that specified in 3.4.11.5. Pressure drop shall then be increased to verify proper operation of the differential pressure indicator.

4.6.15 Cleanability. Both the filter and the QFRE shall be subjected to sufficient contamination so that flow requirements cannot be met. These elements shall then be cleaned by the technical manual procedures specified by the manufacturer to demonstrate cleanability. After cleaning, each element shall be capable of meeting the applicable flow and pressure drop criteria.

4.6.16 Dome integrity (for dome loaded type regulators). Charge dome with 100 percent helium to give mid-range set point. Cycle valve over its full stroke at a rate of approximately five cycles per minute for 8 hours. Measure leakage from dome every 2 hours using a mass spectrometer. Detected leakage shall not exceed a rate which will cause regulated pressure to shift out of the specification performance envelope in a period of less than 1 year. This test is intended to verify ability of the manifold to perform within specification limits over periods of at least 1 year without requiring any adjustment.

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4.6.17 Pressure reversal. The manifold shall be tested to determine the susceptibility to damage when subjected to pressure reversal as specified in 3.4.11.1.4.5. It shall be set up with maximum inlet pressure and maximum set pressure. A separate means shall be included to insure that there is no loss of downstream pressure during this test. Load shall then be removed from the regulator set-point mechanism (if the regulator is spring loaded, back off all spring compression, if it is gas dome loaded, the dome charge shall be released). This condition (no load on the regulator set point mechanism, maximum set pressure applied to the outlet side of the regulator seat) shall be maintained for a period of not less than 1 hour. Inlet pressure shall then be released from the manifold, and this condition (no load on the regulator set point mechanism, zero pressure applied to the inlet side of the regulating valve seat, and maximum set pressure applied to the outlet side of the regulator seat) maintained for an additional period of not less than 1 hour. At the conclusion of the reverse pressure test, the following specific performance checks shall be conducted on the regulator followed by a disassembly of the regulator to examine for damage. There shall be no degradation to performance or evidence of damage to the regulator or any other portion of the manifold.

(a) Accuracy of regulation (see 4.6.4).
(b) Seat-tightness test (see 4.6.9).

4.6.18 Operational. A 5-hour operational test shall be conducted. Prior to start of the 5-hour flow, the regulator and relief valve settings shall be checked and seat-tightness and leakage tests shall be made. During the 5 hours of operation, flow shall be varied from lock-up to maximum rating and back at varying rates. Intermediate flow ranges shall be investigated for signs of instability, icing, inaccurate regulation, element clogging, excessive noise, or other unsatisfactory performance. Line moisture shall be varied within the limits specified in 3.4.12. The 5 hours of operation shall include not less than 25 minutes aggregate time within each of the following specific flow ranges: 95 ± 5 percent maximum rated flow; 50 ± 5 percent maximum rated flow; 25 ± 5 percent maximum rated flow; and 7.5 ± 2.5 percent maximum rated flow. At approximately the mid-point, and at the conclusion of this 5-hour flow test, the following series of specific performance checks shall be conducted:

(a) Accuracy of regulation (see 4.6.4).
(b) Hand valve functioning and operating torque test (see 4.6.7).
(c) Seat-tightness test (see 4.6.9).
(d) External leakage test (see 4.6.10).
(e) Noise (see 4.6.19).

There shall be no adjustments made to the regulators, relief valve or other parts of the manifold during this test. There shall be no evidence of shifts in set-points which would indicate a shift out of the specified performance envelope in a period of 1 year or less. At the conclusion of the 5-hour test, the manifold shall be disassembled and examined for any evidence of excessive wear and impending damage or breakage.

4.6.18.1 Regulating valve cycling. With maximum pressure rating upstream, the regulating valve shall be cycled 1000 times. There shall be no degradation to performance or evidence of damage or excessive wear.

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4.6.19 Noise. The manifolds shall be tested for airborne and structureborne noise in accordance with MIL-STD-740 and the appendix herein. Test points shall be as specified in table III. Acceptance criteria shall be as specified in 3.5.6.

TABLE III. Test points.

Test point no.	Inlet pressure	Set pressure	Flow rate (percent maximum rating)
1	Maximum	Minimum	10
2	Maximum	Minimum	25
3	Maximum	Minimum	50
4	Maximum	Minimum	75
5	Maximum	Minimum	100
6	Maximum	Maximum	10
7	Maximum	Maximum	25
8	Maximum	Maximum	50
9	Maximum	Maximum	75
10	Maximum	Maximum	100
11	Minimum	Maximum	10
12	Minimum	Maximum	25
13	Minimum	Maximum	50
14	Minimum	Maximum	75
15	Minimum	Maximum	100
16	Minimum	Minimum	10
17	Minimum	Minimum	25
18	Minimum	Minimum	50
19	Minimum	Minimum	75
20	Minimum	Minimum	100

4.7 Inspection of packaging. Sample packages and packs, and the inspection of the preservation-packaging, packing and marking for shipment and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

5. PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisition. For the extent of applicability of the packaging requirements of referenced documents listed in section 2, see 6.6.)

5.1 Preservation, packing and marking. Valve manifolds shall be preserved level A or C, packed level A, B or C as specified (see 6.2.1) and marked in accordance with MIL-V-3. Commercial preservation, packing and marking shall be in accordance with the requirements specified by the acquisitioning agency (see 6.2.1 and 6.4).

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5.2 Cushioning and wrapping materials. Use of excelsior, newspaper, shredded paper (all types, including wax paper) and similar hygroscopic or nonneutral materials and all types of loose-fill materials for applications such as cushioning, filler, stuffing and dunnage for materials destined for shipboard stowage and use is prohibited except that vermiculite is approved for packaging applications of liquid (chemical, petroleum, and so forth) products. Cushioning and wrapping materials selected shall incorporate properties and characteristics for resistance to fire, examples are:

UU-P-268	Paper, kraft wrapping type II, grade C or D
PPP-C-850	Polystyrene, expanded grade SE, type I or II
PPP-C-1120	Bound fiber, uncompressed type III or IV
MIL-R-6130	Cellular rubber grade A
MIL-R-20092	Cellular rubber class 1 or 4
MIL-P-26514	Polyurethane foam (rigid or flexible)

5.2.1 Cushioning, filler, dunnage and wrapping materials.

5.2.2 Level A preservation and levels A and B packing. Use of all types of loose-fill materials for preservation and packing applications such as cushioning, filler or dunnage is prohibited for materials destined for shipboard installation or stowage.

5.2.3 Level C and commercial preservation and packing. When loose-fill type materials are used for preservation and packing applications such as cushioning, filler and dunnage, all containers (unit, intermediate and shipping) shall be marked or labeled with the following information:

"CAUTION

Contents cushioned etc., with loose-fill material. Not to be taken aboard ship. Remove and discard loose-fill material. If required, recushion with cellulosic material, bound fiber, fiberboard or transparent flexible cellular material."

5.2.4 Cushioning, filler, dunnage and wrapping materials selected, whenever available, shall exhibit improved performance for resistance to fire.

5.3 Talc and talcum used in the packaging process of items shall be free of asbestos.

6. NOTES

6.1 Intended use. Manifolds specified herein are intended for pressure reducing service in air or nitrogen systems where special acoustic requirements apply.

6.1.1 Manifold selection. Selection of the manifold configuration will be in accordance with shipbuilding specifications. Configuration selection is based on the criticality of the application. The following serves as a guide for selection of manifold configuration:

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(a) Configuration 1-1 (one reducing valve with a manual by-pass) should be selected for applications where, due to the infrequency and slow rate of load variations, the outlet pressure can be effectively regulated with a manual by-pass while the pressure regulating valve or filter is secured for maintenance or repair.
(b) Configuration 2-1 (two reducing valves in parallel) should be selected where it is mandatory to have an alternate source of automatically regulated pressure immediately available if one of the pressure regulators or filters is secured for maintenance or repair.
(c) Configuration SP (special configuration) is included to allow configurations required for special applications. For example, the inclusion of dual relief valve protection, of solenoid controlled loading for the regulators, or a manual by-pass meeting the same acoustic criteria as the automatic regulating valves.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number and date of this specification.
(b) Configuration and pressure rating required (see 1.2 and 1.3).
(c) Whether first article is required (see 3.1).
(d) Size of end connection (see 3.4.3).
(e) Material for union tailpieces (see 3.4.3).
(f) Whether bottom or back mounting is required (see 3.4.4).
(g) Whether a gauge bleed valve is required as part of the manifold (see 3.4.9).
(h) If test and cleaning fixture is other than as specified (see 3.4.14).
(i) Reduced pressure range required (see 3.5.1).
(j) Upstream pressure range (see 3.5.1).
(k) Capacity requirement (see 3.5.1).
(l) Regulated pressure limits, if other than as specified (see 3.5.2).
(m) Relief valve characteristics, if other than as specified (see 3.5.3).
(n) Internal leakage, if other than as specified (see 3.5.4).
(o) Quantity and distribution of manuals (see 3.6).
(p) If test for accuracy of regulation is other than as specified (see 4.6.4).
(q) Levels of preservation and packing required (see 5.1).
(r) Requirements for commercial preservation, packing and marking required (see 5.1 and 5.3).

6.2.2 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL), the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of

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FAR 52.227-7031 are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs.

Paragraph no.	Data requirement title	Applicable DID no.	Option
3.3.2.1.1	Certificate of compliance	DI-E-2121	----
3.6 and 3.6.1	Drawings, engineering and associated list	DI-E-7031	Level 3 Design activity designation - contractor Drawing no. - contractor Delivery of hard copy - contracting activity
3.6 and 3.6.2	Manual, technical, preliminary	DI-M-2043	MIL-M-15071, type I
	Manual, technical, standard	DI-M-2044	MIL-M-15071, type I
4.3.1	First article inspection procedures	DI-T-4901	----
4.3.2	First article inspection report	DI-T-4902	----
4.4	Reports, test	DI-T-2072	10.1.b
60.1.1 of appendix	Report, component shop noise tests	UDI-T-23764	----

(Data item descriptions related to this specification, and identified in section 6 will be approved and listed as such in DoD 5000.19L., Vol. II, AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.2.2.1 The data requirements of 6.2.2 and any task in sections 3, 4, or 5 of this specification required to be performed to meet a data requirement may be waived by the contracting/acquisition activity upon certification by the offeror that identical data were submitted by the offeror and accepted by the Government under a previous contract for identical item acquired to this specification. This does not apply to specific data which may be required for each contract regardless of whether an identical item has been supplied previously (for example, test reports).

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6.2.3 Limited rights legend. When the Government has only limited rights in the data shown on the drawings, as determined by the contractual provisions regarding rights in technical data, the drawings furnished may be marked with the following restrictive legend:

"Furnished under United States Government Contract No. _____. Shall not be either released outside the Government, or used, duplicated, or disclosed in whole or in part for manufacture or procurement, without the written permission of _____, except for: (a) emergency repairs or overhaul work by or for the Government, where the item or process concerned is not otherwise reasonably available to enable timely performance of the work; or (b) release to a foreign government, as the interests of the United States may require; provided that in either case the release, use, duplication or disclosure hereof shall be subject to the foregoing limitations. This legend shall be marked on any reproduction hereof in whole or in part."

6.3 First article inspection. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection as to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Commercial preservation, packing and marking. Guidance for commercial preservation, packing and marking are specified in the joint DOD regulation for Logistics-Packaging of Material-AR700-15, NAVSUPINST 4030.28A, AFG 71-6, MCO 4030.33A, and DLAR 4145.7 appendix A thereto.

6.5 Provisioning. Provisioning Technical Documentation (PTD), spare parts, and repair parts should be furnished as specified in the contract.

6.5.1 When ordering spare parts or repair parts for the equipment covered by this specification, the contract should state that such spare parts and repair parts should meet the same requirements and quality assurance provisions as the parts used in the manufacture of the equipment. Packaging for such parts should also be specified.

6.6 Sub-contracted material and parts. The preparation for delivery requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.7 Supersession data. Requirements for type B valve manifolds in MIL-V-24272A(SHIPS) dated 24 September 1973 are now covered by this specification.

Preparing activity:
Navy - SH
(Project 4820-N482)

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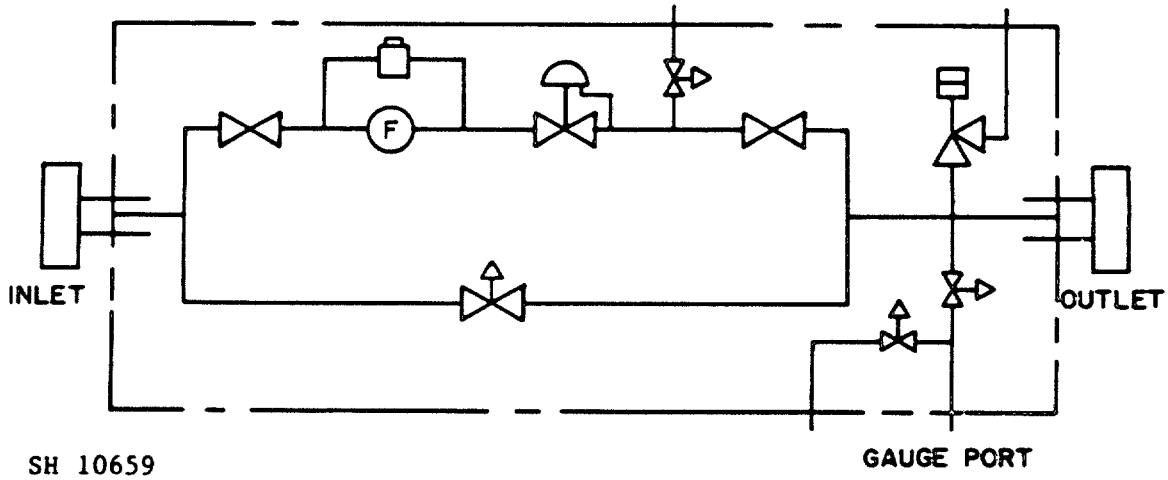


FIGURE 1. Configuration 1-1 manifold.

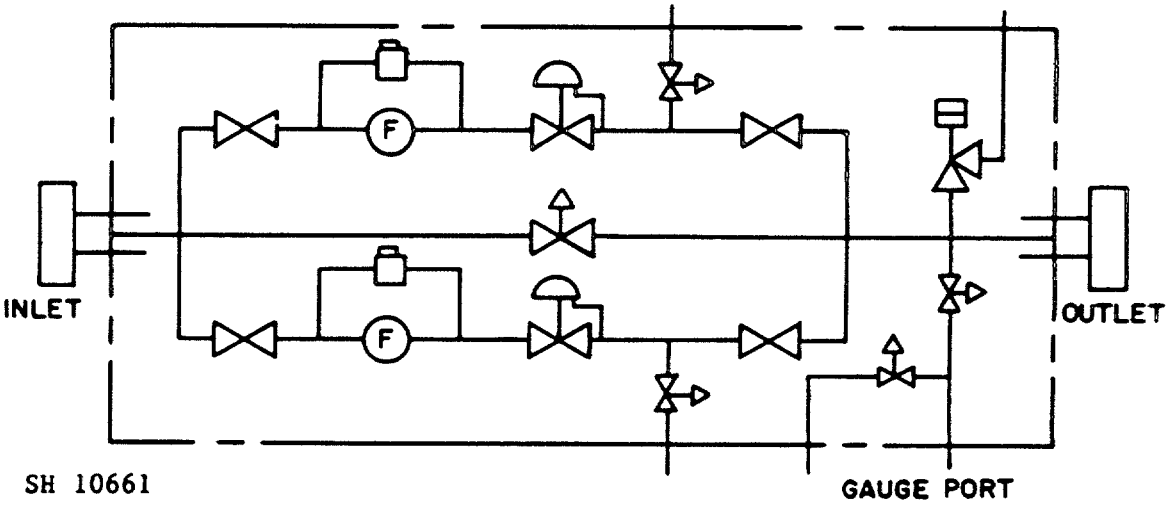


FIGURE 2. Configuration 2-1 manifold.

NOTES TO FIGURES 1 AND 2:

1. The pressure regulators are shown as gas referenced for illustration only and are not necessarily limited to these constructions.

2. Figures 1 and 2 are basic outlines only. Additional vent valves may be necessary for operational and maintenance purposes.

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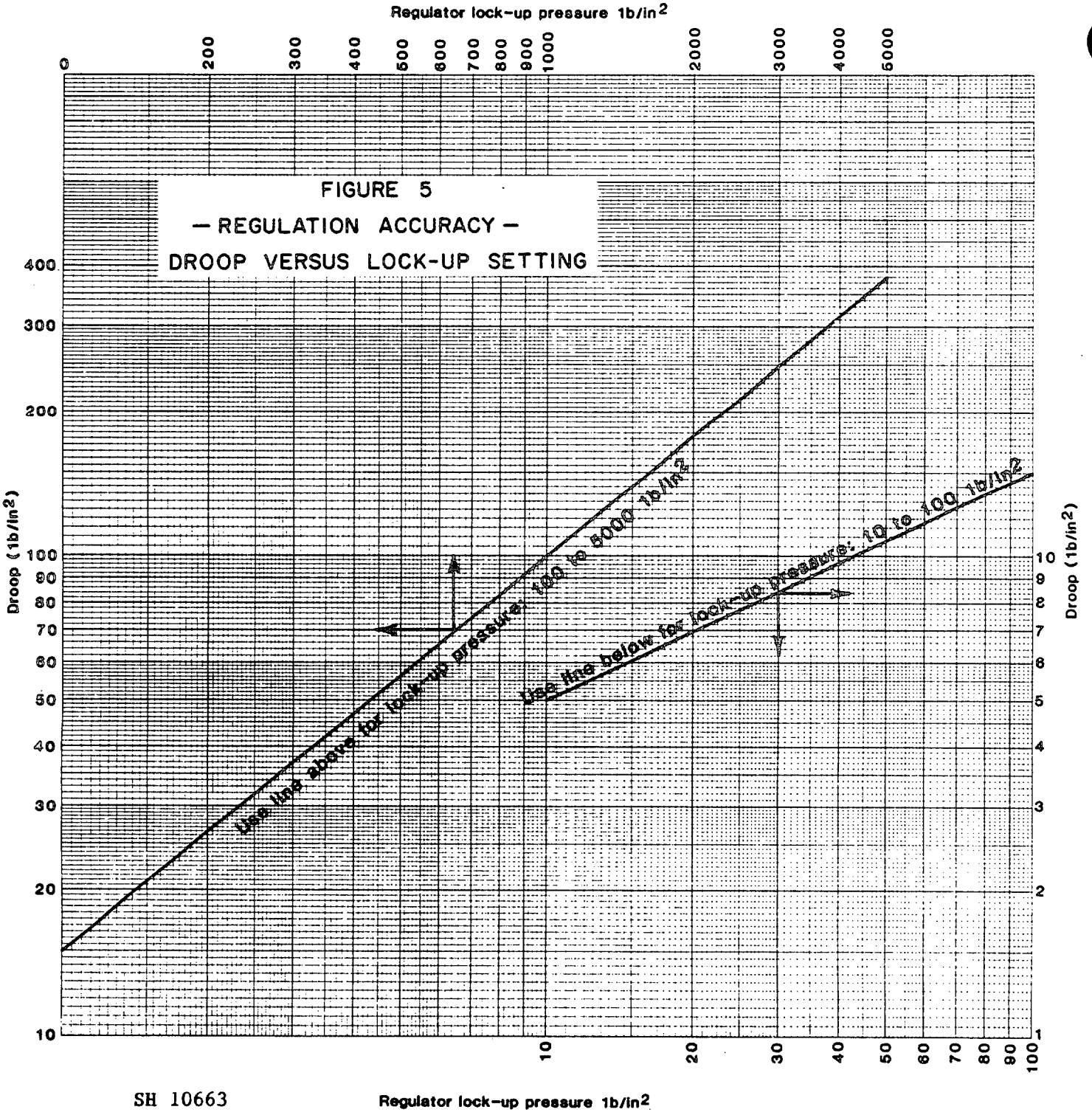


FIGURE 3. Regulation accuracy - droop versus lock-up setting.

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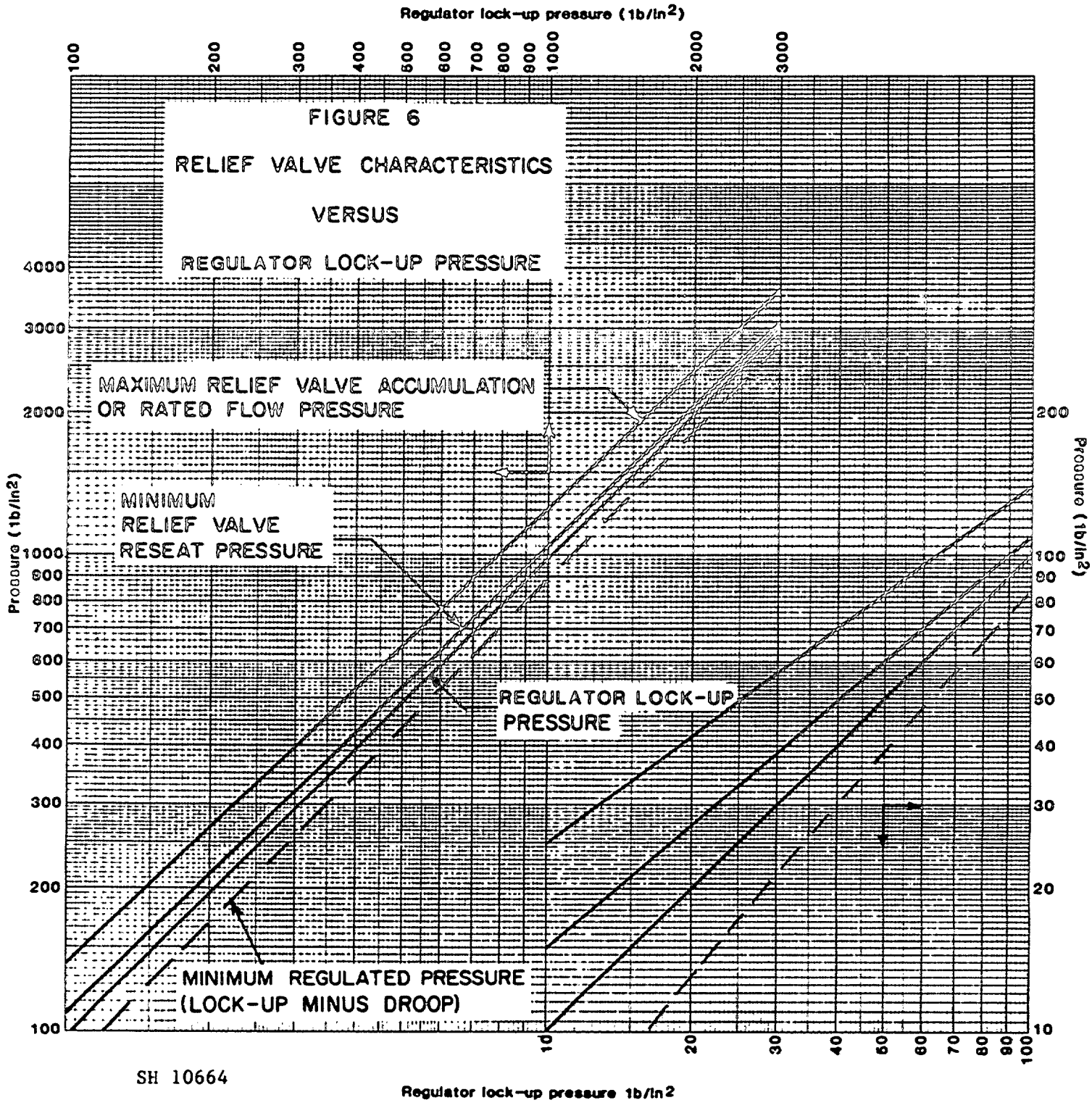
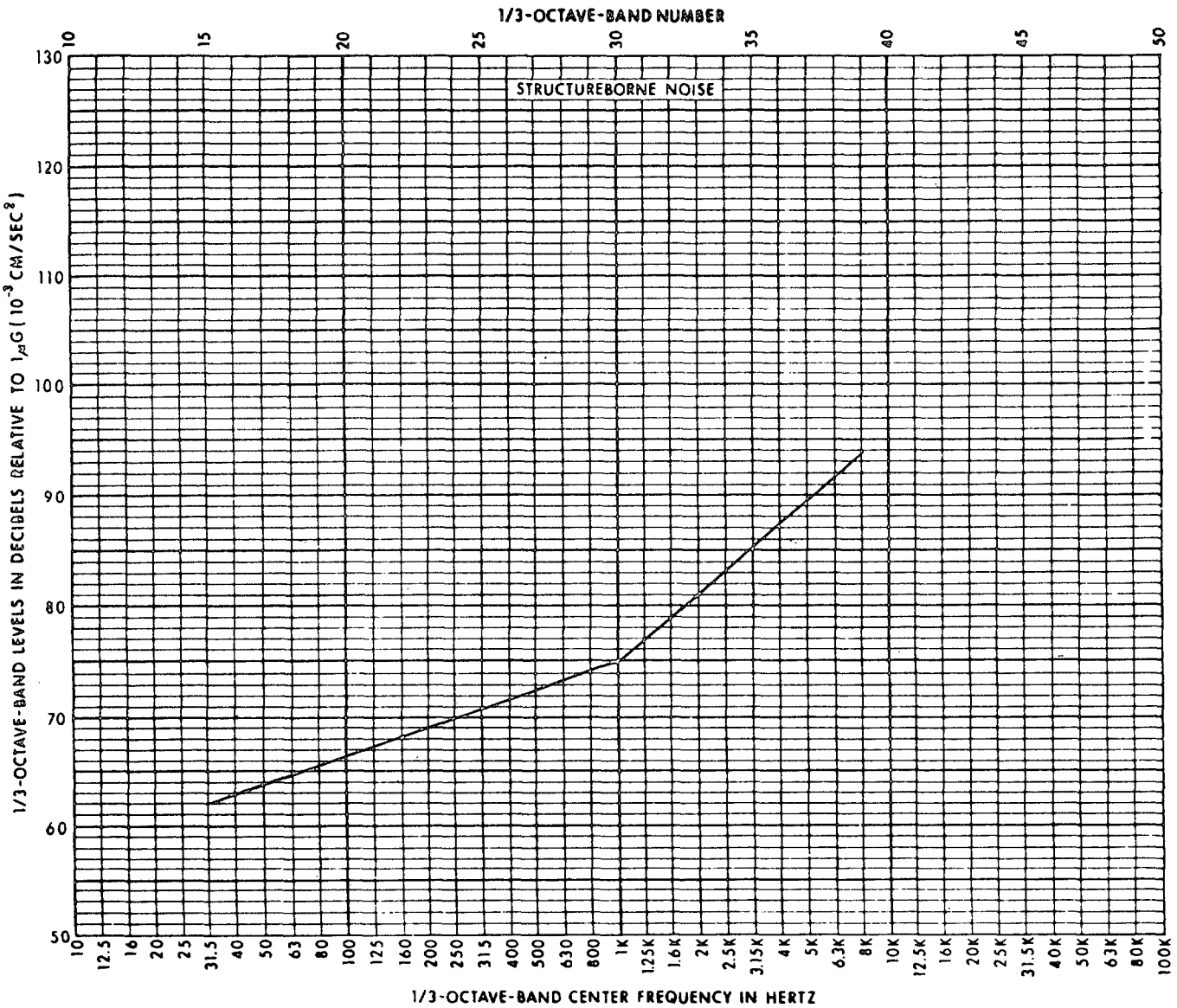


FIGURE 4. Relief valve characteristics versus manifold lock-up pressure.

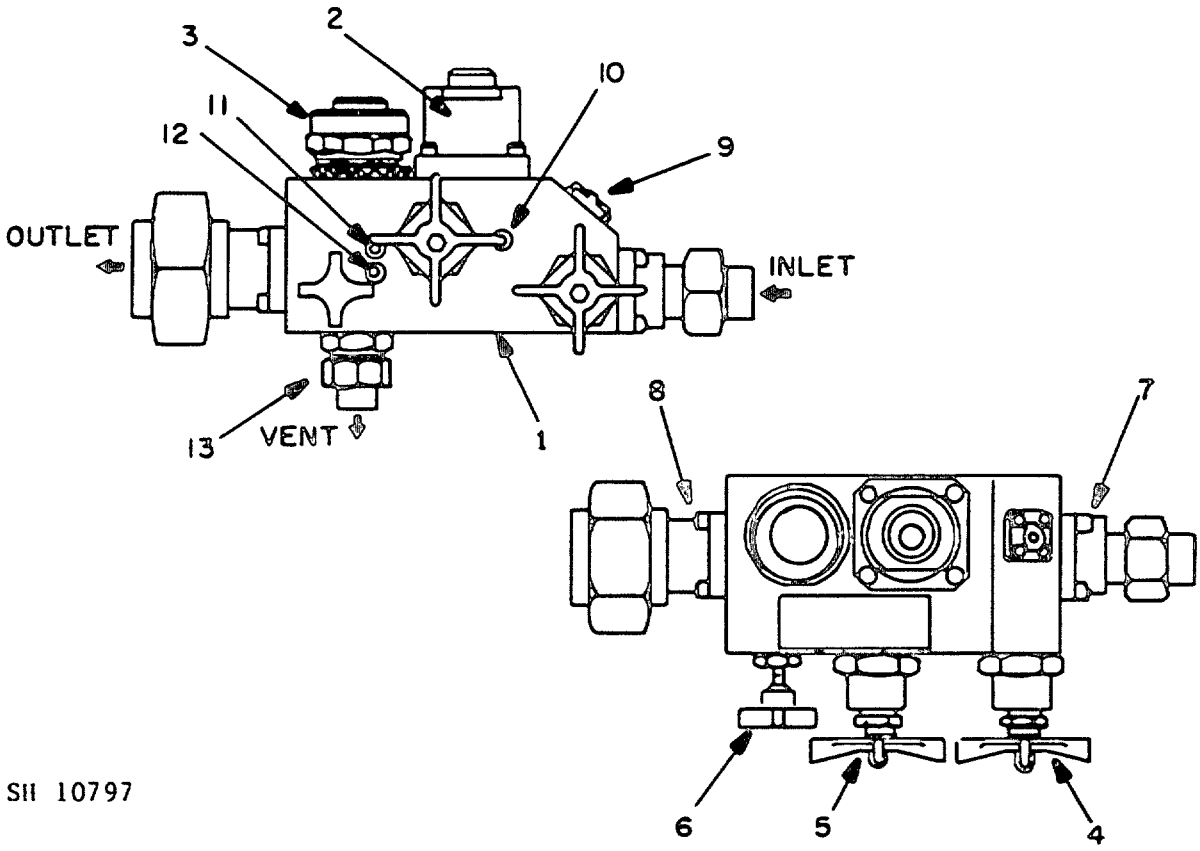
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SH 10665

FIGURE 5. Structureborne noise.

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SH 10797

Manifold subassembly index			
Item no.	Part no.	Dwg. no.	Nomenclature
1	6851	70712	Manifold block assembly
2	7781	31168	Regulating valve subassembly
3	2638	66776	Relief valve subassembly
4	4320	22261	Inlet stop valve subassembly
5	4350	31186	Outlet stop valve subassembly
6	3031	24381	By-pass valve subassembly
7	4198	61617	Inlet fitting subassembly
8	4895	88876	Outlet fitting subassembly
9	2113	11191	Filter subassembly
10	4227	38222	Filter bleed subassembly
11	8325	67241	Gauge bleed subassembly
12	6724	66766	Gauge shut-off subassembly
13	2904	15422	Relief outlet fitting subassembly
--	----	26001CD	C.D. sheet
--	----	72959	Special tool table
--	----	32324	Manifold assembly installation
--	----	10122	Table of general notes

Pressure-regulating manifold	Manufacturer's name and address.
Drawing No. 2351	Part no. 6061

FIGURE 6. Manifold drawing layout.

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APPENDIX

STRUCTUREBORNE AND AIRBORNE NOISE TESTING DETAILS

10. SCOPE

10.1 Scope. This appendix details noise testing for quiet design valves.

20. APPLICABLE DOCUMENTS

20.1 Government documents. The following documents form a part of this appendix to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-M-17185 - Mounts, Resilient; General Specifications and Tests for (Shipboard Application).
MIL-M-17508 - Mounts, Resilient: Types 6E2000, 6E900, 6E900BB, 7E450, 7E450BB, 6E150, and 6E100.
MIL-M-19379 - Mounts, Resilient, Mare Island Types 11M15, 11M25, and 10M50.
MIL-M-19863 - Mounts, Resilient: Type 5B5000H.
MIL-M-21649 - Mount, Resilient; Type 5M10,000-H.

STANDARD

MILITARY

MIL-STD-740 - Airborne and Structureborne Noise Measurements and Acceptance Criteria of Shipboard Equipment.

DRAWING

NAVAL SEA SYSTEMS COMMAND (NAVSEA)

NAVSHIPS 810-2145600 - Mounts, Resilient, Unbonded, Spool, Portsmouth Type.

PUBLICATION

NAVAL SEA SYSTEMS COMMAND

0939-LP-001-2010 - A Guide to Structural Damping for Naval Applications.

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

30. QUIET VALVE TEST FACILITY

30.1 The test facility shall be capable of reproducing the shipboard operational conditions for the valve being tested. Operational parameters such as pressure, temperature, flow rate, flow velocity, and so forth, shall be as specified in this specification. The valve shall be connected between

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straight pipe of appropriate schedule and nominal diameter for which the valve pipe connections were designed. In order to minimize system noise influence on the valve and to provide a standardized basis for comparison, the straight pipe sections shall be at least 10 pipe diameters in length and shall be damped in accordance with Navy pipe damping procedures as specified in NAVSEA 0939-LP-001-2010 for at least 10 pipe diameters. In order to reduce system noise, and when conditions permit their use, flexible hoses, acoustic filters or other noise attenuating or isolating devices may be incorporated in the piping system at distances greater than 10 pipe diameters from the valve. System background noise should be at least 10 decibels below the valve's acceptance criteria. Piping needing support within 10 pipe diameters of the valve being tested shall be isolation mounted from its supports. Generally, low frequency mounts shall be used in accordance with MIL-M-17508, MIL-M-19379, MIL-M-19863, MIL-M-21649 and Drawing 810-2145600. Where other than Navy standard mounts are used they shall be in accordance with MIL-M-17185. However, distributed isolation material, such as Isomode or equal, if considered sufficient to eliminate influence of the test facility from background noise sources, may be used.

40. VALVE TRANSDUCER LOCATIONS

40.1 Transducer locations shall be selected so as to determine as closely as possible the vibratory excitation that would be transmitted by the valve to the ship either through the piping system or foundation. For those valves which are supported only by piping, the transducers shall be located on the appropriate inlet and outlet connections of the valve. Structureborne noise measurements shall be made at these locations in directions perpendicular and parallel to the flow at the piping connections. For valves which are also supported by foundations, additional transducer locations at the attachment points (feet) of the valve are required. Transducers shall be located on the base of the valve at two diagonally opposite corners above mounts and as close as possible to the valve's designed attachment points. At least two mutually perpendicular measurement directions are required for each location.

50. QUIET VALVES FOR TEST

50.1 In all cases the quiet valve shall be oriented in its normal operating position and supported by its designed piping attachment points directly to the hard piping without flexible pipe connections. Foundations and supports shall be provided for those valves which, because of the size and weight, require same in shipboard systems. These valves shall be isolation mounted to their foundations during the noise test.

60. NOISE TEST SAMPLING PLAN

60.1 This plan applies to each production lot, or portion thereof, of manifolds acquired under a contract which are of the same configuration, design features, rating and intended shipboard application. Testing shall be initiated with step A. If and when step B is achieved, testing shall continue under step B until all manifolds are completed, or until reversion back to step A is made necessary by the occurrence of a failure under step B. If reversion to step A occurs, re-entry into step B shall be achieved upon successful completion of step A.

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60.1.1 A noise test report shall be prepared in accordance with MIL-STD-740 for each manifold tested under this sampling plan (see 6.2.2).

60.2 Step A. Test the first three manifolds from the production lot, or remaining in the production lot.

60.2.1 If no failures occur, proceed to step B.

60.2.2 If any of the three sample lot manifolds fails to meet noise requirements, or any other contract requirement as revealed by this testing, repair and successfully retest the discrepant manifold. A detailed description of the cause of the excessive noise, other discrepant performance and the correction technique used, shall form a part of the noise test report for these manifolds. No design or material deviation shall be made to approved drawings or manifolds without Government approval. Subsequent to repair and retest, required by either noise or other major contract requirement discrepancy, take the next three manifolds from the production lot, and conduct another step A.

60.3 Step B. Test one manifold chosen randomly from each five manifolds, or part thereof, remaining in the production lot. If and when a manifold fails to meet noise requirements, or any other contract requirement as revealed by this testing, repair and successfully retest the discrepant manifold. A detailed description of the cause of the excessive noise, other discrepant performance and the correction technique used, shall form a part of the noise test report for these manifolds. No design or material deviation shall be made to approved drawings or manifolds without Government approval subsequent to repair and retest, required by either noise or other major contract requirement discrepancy, testing shall revert to step A.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-V-24658(SH)	

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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